



SIDDHARTHA KOSTI

Generative AI LLMOps

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Projects

- **Ragaai-catalyst: LLM Management Tool**
 - ♦ Developed and optimized an Evaluation module for LLM metric monitoring and performance analysis.
 - ♦ Created the Prompt Slug module for efficient prompt management across various projects and teams.
 - ♦ Built a synthetic data generation module to produce diverse and realistic data from original sources.
 - ♦ Contributed to Tracer tool for logging trace records.
- **LLM Pipeline Evaluation Tests**
 - ♦ Designed and conducted tests for hallucination and faithfulness.
 - ♦ Created contextual relevancy and readability tests for LLM outputs.
 - ♦ Built tests for refusal/false refusal and detection of prompt-injection, PII, toxicity, bias, and secrets.
- **LLM RAG Chatbot for Hospital System**
 - ♦ Technologies used: LangChain, Neo4j AuraDB, Docker, Streamlit, LiteLLM, Prompt Engineering.
 - ♦ Integrated LangChain and Neo4j AuraDB for data processing and storage.
 - ♦ Developed Agents and customized prompts for text-to-SQL functionality.
 - ♦ Dockerized the application for scalability and deployment.
 - ♦ Utilized litellm and groq-API for cost elimination.
 - ♦ Designed a user interface with Streamlit.
- **Golden Dataset & Auto-Prompt Pipeline**
 - ♦ Build golden dataset for LLM pipeline benchmarking.
 - ♦ Implementing auto-prompt optimization pipeline.
- **Custom Model Training**
 - ♦ Utilized transfer learning and fine-tuning to enhance performance for client applications.
 - ♦ Used Active Learning approach to obtain similar performance with smaller dataset.
 - ♦ Examined TP, FP and FN, Failure Model analysis.
 - ♦ Conducted Failure Mode Analysis to compared two different models performance.
- **Clustering, Outlier, Drift & Labelling Tests**
 - ♦ Utilized K-Means to cluster train+test dataset to find similar data points, outliers and drift if exists.
 - ♦ Utilized SAM (Segment Anything Model) to find labeling consistency issues.

Experience

- **Data Analyst: RagaAI Pvt Ltd**
06/2023 - Present
 - ♦ Developed a claim generation pipeline for LLM as a Judge based metric evaluation.
 - ♦ Implemented PII/Toxicity/Bias/NSFW Guardrails on LLM-generated input and output data.
 - ♦ Implemented LLM judge metrics to evaluate chatbots responses.
 - ♦ Refined the hallucination detection prompts for improved accuracy and performance.
 - ♦ Created interactive PoC streamlit dashboard.
 - ♦ Designed and executed data conversion and sanity-check pipelines.
- **Data Analyst: Affy Informatics Pvt Ltd**
10/2021 - 05/2023
 - ♦ Developed RAG chatbot focused on NFT blockchain data specific to contract addresses.
 - ♦ Analyzed blockchain data using various open-source platforms.
 - ♦ Built tailored dashboards for client use.
- **Research Experience**
06/2014 - 09/2021
 - ♦ Over seven years of experience teaching AI/ML courses at the undergraduate level.
 - ♦ Published manuscripts in reputed SCI journals.

Education

- **M.Tech: IIT Kanpur** **07/2011 – 7/2013**
- **B.E.: MITS Gwalior** **07/2006 – 6/2010**

Tools & Technologies

- LLM: OpenAI, Huggingface, Langfuse, LangChain, LiteLLM, Groq, Claude
- Deployment: AWS, Docker
- Tracking: Git, GitHub
- Dashboard: streamlit
- Other: TensorFlow, PyTorch

Achievements

- Secured two research grants totaling ₹22.36 lakhs to work on Solar PV-T Panel projects.
- Achieved AIR 677 in GATE 2011.